

LAWRENCE CHILLRUD

Ph.D. Candidate

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 CV Date of Preparation: 9/2025

Education

- 9/2022 – Present **Ph.D. in Electrical Engineering** *Evanston, IL*
Northwestern University, McCormick School of Engineering
Advisors: [Aggelos Katsaggelos, Ph.D.](#) and [Lee Cooper, Ph.D.](#)
Specialization in Signals and Systems
- 9/2021 – 5/2022 **Post-baccalaureate Scholar** *New York, NY*
Columbia University, School of Professional Studies
- 9/2016 – 5/2020 **B.A. in Computer Science** *New York, NY*
Columbia University, Columbia College
Specialization in Intelligent Systems

Awards and Honors

- 9/2024 – Present **National Science Foundation Graduate Research Fellowship**
\$159,000 in funding over three years
[Awarded](#) to support education of “outstanding graduate students... who have demonstrated their potential for significant research achievements in STEM.” Proposal entitled *Closing the gap b/w deep ensembles & Bayesian inference in medical imaging*.

Research

- 9/2022 – Present **Graduate Student Research Fellow** *Chicago, IL*
Northwestern University, Feinberg School of Medicine
Department of Pathology
Advisor: [Lee Cooper, Ph.D.](#)

Working in the [Computational and Integrative Pathology Group](#) to develop ML methods to analyze H&E stained histopathology whole slide imaging data. Work includes:

- Close collaboration & weekly presentation of work & experimental results w/pathologists
- Weakly-supervised/multiple instance learning for a) Banff lesion score prediction from renal allograft biopsies and b) survival analysis of a cohort of breast cancer patients
- Investigating DL/CNN digital pathology models using state-of-the-art transformer architectures, self-supervised foundation models
- Working with HistomicsTK to clean, manage & query massive digital pathology datasets
- Experiments investigating a) foundation model sensitivity to types of stains used in whole slide images, and b) efficacy of stain normalization approaches
- Dimension reduction algos. to visualize low dim. embedding spaces of learned features
- Developing novel loss function to minimize false positives at a high true positive rate
- Software dev. on glimr API, a simplified wrapper for hyperparam. search w/Ray Tune

9/2022 – Present

Graduate Student Research Fellow

Evanston, IL

Northwestern University, McCormick School of Engineering

Department of Electrical & Computer Engineering

Advisor: [Aggelos Katsaggelos, Ph.D.](#)

Working in the [Image and Video Processing Lab](#) to develop ML models to predict Meningioma/Glioblastoma biomarker statuses from a patient's mpMRI, entailing:

- Close collaboration & weekly presentation of work & experimental results w/radiologists
- Semi-supervised deep learning w/convolutional networks (e.g. U-Nets, transformers)
- Uncertainty quantification experiments w/Bayesian neural networks & deep ensembles
- Explanatory modeling w/rigorous double nested leave-two-out cross-validation approach to train lasso logistic regression models to learn from limited cohort size
- Texture analysis study using custom-extracted PyRadiomics & CoLIAGe features
- Design & validation of preprocessing pipeline to prepare MRI scans in the cohort for analysis, including but not limited to: scan type cleanup, DICOM to NIfTI conversion, N4 bias field correction, skullstripping, intensity standardization, & image registration
- Other projects include active learning to inform the sampling of collocation points in physics informed neural networks modeling binary star system evolution dynamics

6/2025 – 8/2025

Machine Learning & Digital Biomarker Intern

Cambridge, MA

Takeda Pharmaceuticals

Oncology Precision & Translational Medicine

Manager: [Melinda Chen, Ph.D.](#)

Worked as a summer intern to expand Takeda Oncology's digital pathology capabilities in clinical trials. Contributions included:

- Authored in-house python package examining quantitative relationships b/w cells & tissue structures in H&E, IHC images, enhancing tumor microenvironment characterization
- Assessed feasibility, performance, and translational value of virtual-staining models for biomarker discovery, spatial profiling, and clinical trial enrichment, designing pilot study
- Wrote comparative landscape analysis, lecture series on digital path foundation models

10/2020 – 8/2022

Senior Programmer

New York, NY

Columbia University, Mailman School of Public Health

Department of Environmental Health Sciences

Advisor: [Marianthi-Anna Kioumourtzoglou, Sc.D.](#)

Worked in [makLab](#) to develop interpretable machine learning methods for assessing complex mixtures of environmental exposures. Main project tailored Principal Component Pursuit (PCP), a dimensionality reduction technique from computer vision, for pattern recognition in environmental epidemiology, including development of R software package. Other work included:

- Investigated convex & non-convex approaches to matrix decomposition, dim reduction
- Leveraged Gaussian processes to design faster cross-validated grid searches
- Developed Bayesian non-parametric ensemble model for uncertainty characterization
- Conducted extensive R code reviews for academic papers (reviewed over 9,000 lines)
- Cleaned, visualized & documented public health datasets for various research questions
- Aided in writing, editing of scientific papers & abstracts, presented work at conferences
- Explored methods for source apportionment: PCP, PCA, Autoencoders, Factor Analysis
- Designed, implemented, evaluated environmental/epidemiological health models/analyses

6/2020 – 10/2020

EHS Research Assistant

New York, NY

Columbia University, Mailman School of Public Health

Department of Environmental Health Sciences

Advisor: [Marianthi-Anna Kioumourtzoglou, Sc.D.](#)

Worked in [makLab](#) to develop interpretable machine learning methods for assessing complex mixtures of environmental exposures:

- Adapted & extended Principal Component Pursuit for environmental mixtures data
- Benchmarked PCP's computational efficiency, interrogated its mathematical foundations
- Developed novel, user-friendly R packages for implementation of environmental PCP
- Designed & ran synthetic & applied experiments to assess PCP's statistical performance

6/2020 – 10/2020

NLP Research Assistant

New York, NY

Columbia University

Department of Computer Science

Advisor: [Kathleen McKeown, Ph.D.](#)

Worked to develop, train, and validate an automatic fact-checking model for combating misinformation online surrounding COVID-19 & climate-change:

- Researched transformer architectures (BERT), few-shot learning, claim detection, named entity recognition, unsupervised data augmentation, transfer learning for fact-checking
- Built a COVID-19-specific dataset to train and test RoBERTa-based fact-checking model
- Scraped millions of online news articles for COVID-19 claims, mapped to scientific papers
- Wrote IRB protocol to receive approval for human annotators to tag fact-checking dataset
- Implemented & maintained user-friendly annotation interface to facilitate annotations
- Assisted in writing, editing of scientific paper detailing our novel fact-checking pipeline

Publications

Peer-Reviewed

12. Benavides J, Cohen JW, **Chillrud LG**, Cohen G, DeSerisy M, Ramphal B, Goldsmith J, Kioumourtzoglou M-A, Margolis AE. [Prenatal environmental risk for adolescent attention, thought, and substance use problems: A mixtures approach to coexposure and psychiatric comorbidities](#). *Environmental Epidemiology*, 2025 Oct 1;9(5):e417.
11. Ayad MA, Nateghi R, Sharma A, **Chillrud LG**, Seesillapachai T, Chou T, Cooper LAD, Goldstein JA. [Deep learning for fetal inflammatory response diagnosis in the umbilical cord](#). *Placenta*, 2025.
10. Benavides J, Carrillo-Gallegos C, Kumar V, Rowland ST, **Chillrud LG**, Adeyeye T, Paisley J, Coull B, Henze DK, Martin RV, Fiore AM, Kioumourtzoglou M-A. [bneR: A collaborative workflow for air pollution exposure modeling and uncertainty characterization using the Bayesian Nonparametric Ensemble](#). *Journal of Environmental Management*, 375 (2025): 124061.
9. Wu H, Kalia V, Manz KE, **Chillrud LG**, Dishon NH, Jackson GL, Dye CK, Orvieto R, Aizer A, Levine H, Kioumourtzoglou M-A, Pennell KD, Baccarelli AA, and Machtinger R. [Exposome Profiling of Environmental Pollutants in Seminal Plasma and Novel Associations with Semen Parameters](#). *Environmental Science & Technology*, 2024 58 (31), 13594-13604.
8. Wang G, Harwood KR, **Chillrud LG**, Ananthram A, Subbiah M, McKeown KR. [Check-COVID: A Corpus and Task for Fact-Checking COVID-19 Misinformation with Scientific](#)

Evidence. *Findings of the Association for Computational Linguistics: ACL 2023*, pages 14114–14127, Toronto, Canada. Association for Computational Linguistics.

7. Cerna-Turoff I, **Chillrud LG**, Rudolph KE, Casey JA. [Standards in responsibly sharing cohort data for transparency and reproducibility: response to The Young Lives study](#). *International Journal of Epidemiology*, 2023 Oct 5;52(5):1666-1669.
6. Tao RH, **Chillrud LG**, Nunez Y, Rowland ST, Boehme AK, Yan J, Goldsmith J, Kioumourtzoglou M-A. [Applying Principal Component Pursuit to investigate the association between source-specific fine particulate matter and myocardial infarction hospitalizations in New York City](#). *Environmental Epidemiology* 7(2):p e243, April 2023.
5. Gibson EA, Zhang J, Yan J, **Chillrud LG**, Benavides JP, Nunez Y, Herbstman JB, Goldsmith J, Wright J, Kioumourtzoglou M-A. [Principal Component Pursuit for Pattern Identification in Environmental Mixtures](#). *Environmental Health Perspectives*, 2022.
4. Rowland ST, **Chillrud LG**, Boehme AK, Wilson A, Rush J, Just AC, Kioumourtzoglou M-A. [Can Weather Help Explain 'Why Now?': The Potential Role of Hourly Temperature as a Stroke Trigger](#). *Environmental Research*. 2022 May 1;207:112229.

Workshop

3. Wang G, **Chillrud LG**, McKeown KR. [Evidence based Automatic Fact-Checking for Climate Change Misinformation](#). *SocialSens Workshop on The International AAAI Conference on Web and Social Media*, 2021.

Software packages

2. **Chillrud LG**, Benavides J, Gibson EA, Zhang J, Yan J, Wright JN, Goldsmith J, Kioumourtzoglou M-A. [pcpr: Principal Component Pursuit for Environmental Epidemiology](#). *CRAN R package version 1.0.0*, 2025.
1. McBrien H, Casey JA, **Chillrud LG**, Flores NM, Wilner LB. [popexposure: Find the number of people residing near environmental hazards quickly and efficiently](#). *Python package version 1.0.0*, 2025.

Presentations

Oral

5. **Chillrud LG**, Gibson EA, Nunez Y, Colgan R, Tao RH, Zhang J, Yan J, Wright J, Goldsmith J, Kioumourtzoglou M-A. [Principal Component Pursuit for Pattern Recognition from Incomplete Environmental Data](#). *ENAR 2022*, Houston, TX, March 27-30, 2022.
4. Benavides JP, Nunez Y, **Chillrud LG**, Gibson EA, Kioumourtzoglou M-A. [Pre- and Postnatal Urban Exposure Patterns and Childhood Neurobehavior](#). *Exposome Data Challenge*, ISGlobal, April 28-30, 2021.

Invited Talks

3. **Chillrud LG**. [Parallel computation in R with the foreach package: A brief introduction](#). *RClub*, Columbia Mailman School of Public Health, New York, NY, April 7, 2022.

Posters

2. **Chillrud LG**, Yan J, Wright J, Goldsmith J, Kioumourtzoglou M-A. [Principal Component Pursuit for Source Apportionment from Block Missing Data: An Application to NYC PM_{2.5} Data](#). *ISEE 2022*, Athens, Greece, September 18-21, 2022.
1. **Chillrud LG**, Gibson EA, Nunez Y, Colgan R, Tao RH, Zhang J, Yan J, Wright J, Goldsmith J, Kioumourtzoglou M-A. [Principal Component Pursuit for Exposure Pattern Recognition: An Application to Persistent Organic Pollutants and Leukocyte Telomere Length](#). *ISEE 2021*, New York, NY, August 23-26, 2021.

Technical Skills

<i>Languages:</i>	Python, R, $\text{\LaTeX}$, C, C++, Java, HTML, CSS, MATLAB, Bash, Zsh
<i>Operating Sys:</i>	UNIX, macOS
<i>VC Systems:</i>	Git, GitHub
<i>Databases:</i>	MongoDB, NoSQL
<i>Certifications:</i>	HIPAA, CITI, Human Subjects Protection, Responsible Conduct of Research
<i>ML Libraries:</i>	TensorFlow, Keras, PyTorch, Scikit-learn, Hugging Face Transformers, NumPy, SciPy, Pandas, Matplotlib, Seaborn
<i>Dev Tools:</i>	Visual Studio Code, iTerm2, Vim, GNU Screen, tmux, RStudio, Jupyter Notebook, Google Cloud Platform, Conda, Docker

Relevant Coursework

Courses marked **in bold** indicate Graduate level coursework

<i>Computer Sci:</i>	Machine Learning for Medical Images & Signals, Deep Learning for Medical Image Analysis, Machine Learning, Deep Learning, Statistical Pattern Recognition, Probabilistic Graphical Models, Deep Reinforcement Learning, Seminar in Reliable Machine Learning Theory, HCI Methods for Health Research, Explanation & Reproducibility in Data Driven Science, Computational Genomics, Natural Language Processing, Artificial Intelligence, Analysis of Algorithms , Causal Inference, Computer Science Theory, Advanced Programming in C, Computer Systems, Data Structures & Algorithms
<i>Mathematics:</i>	Random Processes, Probability & Statistics , Foundations of Linear Optimization, Linear Algebra, Calculus I, II, & III, Discrete Mathematics, Number Theory, Cryptography

References

References are available upon request. Academic transcripts are available upon request.